

EX280V4.18K Questions

General Instructions:

Duration: 3 hours

You will have the following VMs in the exam

- loadbalancer.domainX.example.com (where X is a variable factor)
- registry.domainX.example.com
- workbench.domainX.example.com (Where we will do our work)
- master.domainX.example.com
- storage.domainX.example.com

Cluster is already configured and all required packages are installed. In case you need any package, you can install with preconfigured repositories.

Docs are available in html and pdf format.

Cluster Web Console and API url will be given in the exam:

<https://console.domainX.example.com:6443> <https://api.domainX.example.com:6443>

After starting the exam first step is to login into the cluster:

kubeadmin user password is stored in /root/kubeadmin-passwd file on workbench VM.

After entering in exam, open a terminal and do ssh on the workbench VM by given password in the information section (one common password will be given in the exam)

View the password stored here and then login into the cluster.

- `cat /root/kubeadmin-passwd labishub`
- `oc login -u kubeadmin -p <password> <cluster URL>`

Q. 1. Configure an Identity Provider

Configure your Openshift Cluster to use an HTPasswd identity provider with the following.

The name of the Identity Provider is : ex280-htpasswd

The name of the secret is : ex280-idp-secret

The user account armstrong is present and can login with password indionce

The user account collins is present and can login with password veraster

The user account aldrin is present and can login with password roonkere

The user account jobs is present and can login with password sestiver

The user account wozniak is present and can login with password glegunge

If the httpd-tool... package is not installed, then use below command

yum install httpd-tool* -y , yum provides http

`htpasswd -c -b -B ex280-htpasswd armstrong indionce`

`htpasswd -b ex280-htpasswd collins veraster`

```
htpasswd -b ex280-htpasswd aldrin roonkere
```

```
htpasswd -b ex280-htpasswd jobs sestiver
```

```
htpasswd -b ex280-htpasswd wozniak glegunge
```

```
oc create secret generic ex280-idp-secret --from-file=htpasswd=ex280-htpasswd -n openshift-config
```

```
$oc get oauth cluster -o yaml > oauth.yaml
```

```
$vim oauth.yaml
```

```
spec:
```

```
identityProviders:
```

```
- htpasswd:
```

```
    fileData:
```

```
        name: ex280-idp-secret
```

```
    mappingMethod: claim
```

```
    name: ex280-htpasswd
```

```
    type: HTPasswd
```

```
:wq
```

```
$oc apply -f oauth.yaml
```

```
$watch oc get pod -n openshift-authentication
```

Q. 2: Configure Cluster Permission.

Configure your openshift cluster to meet the following requirements:

The user account jobs can perform cluster administration tasks.

The user account wozniak can create projects.

The user account wozniak cannot perform cluster administration tasks.

The user account armstrong cannot create projects.

The user account kubeadmin is not present.

Solution:

```
oc adm policy add-cluster-role-to-user cluster-admin jobs
```

```
oc get clusterrolebinding -o wide |grep self
```

```
oc adm policy remove-cluster-role-from-group self-provisioner system:authenticated:oauth
```

```
oc adm policy add-cluster-role-to-user self-provisioner wozniak
```

oc login -u jobs -p indionce ;oc get nodes; oc whoami ; oc whoami --show-console; get the url and open the gui with jobs; then try to login each user which was created earlier and login jobs again and If everything is fine then go below

```
oc delete secret kubeadmin -n kube-system
```

Q. 3. Configure project permissions.

Configure your openshift cluster to meet the following requirements:

The following projects exist: apollo

manhattan gemini bluebook

titan

The user account armstrong is an administrator for project apollo and project gemini.

The user account wozniak can view project titan but not administer or delete it.

Solution:

```
oc new-project apollo
```

```
oc new-project manhattan
```

```
oc new-project gemini
```

```
oc new-project bluebook
```

```
oc new-project titan
```

```
oc policy add-role-to-user edit armstrong -n apollo
```

```
oc policy add-role-to-user edit armstrong -n gemini
```

```
oc policy add-role-to-user view wozniak -n titan
```

Q. 4: Configure Groups:

Configure your OpenShift cluster to meet the following requirements:

The user account armstrong is a member of the commander group.

The user account collins is a member of the pilot group.

The user account aldrin is a member of the pilot group.

Member of the commander group have edit permission in the apollo project.

Member of the pilot group have view permission in the apollo project.

Solution:

```
oc adm groups new commander
```

```
oc adm groups new pilot
```

```
oc adm groups add-users commander armstrong
```

```
oc adm groups add-users pilot collins
```

```
oc adm groups add-users pilot aldrin
```

```
oc policy add-role-to-group edit commander -n apollo
```

```
oc policy add-role-to-group view pilot -n apollo
```

Q5 Create Limit Range for project manhattan:

Set the pod memory limit between 5Mi and 300Mi. - Set the container memory limit between 5Mi and 300Mi and container default request limit for memory is 100Mi.

Set the pod cpu limit between 5m and 300m. Set the container cpu limit between 5m and 300m and container default request limit for cpu is 100m.

```
[student@workstation ~]$ cat limitrange.yml
apiVersion: v1
kind: LimitRange
metadata:
  name: mem-limit-range
  namespace: manhattan
spec:
  limits:
    - type: Pod
      max:
        memory: 300Mi
        cpu: 300m
      min:
        memory: 5Mi
        cpu: 5m
    - type: Container
      max:
        memory: 300Mi
        cpu: 300m
      min:
        memory: 5Mi
        cpu: 5m
      defaultRequest:
        memory: 100Mi
        cpu: 100m
[student@workstation ~]$
```

```
[student@workstation ~]$ oc get limitrange
NAME                               CREATED AT
mem-limit-range                    2026-03-14T07:01:33Z
[student@workstation ~]$ oc describe limitranges mem-limit-range
Name:          mem-limit-range
Namespace:     manhattan
Type           Resource  Min  Max    Default Request  Default Limit  Max Limit/Request Ratio
----
Pod            cpu       5m   300m   -               -               -
Pod            memory    5Mi  300Mi  -               -               -
Container      cpu       5m   300m   100m            300m            -
Container      memory    5Mi  300Mi  100Mi           300Mi           -
[student@workstation ~]$
```

Q. 6. Configure a secure route

Configure the oxcart application in the area51 project with the following requirements:

Complete and Continue The application uses a secure route called oxcart Traffic between the client and the router is encrypted Traffic between the router and the service is unencrypted

The route uses a CA signed certificate with the following subject fields:

/C=US/ST=NV/L=Hiko/O=CIA/OU=USAF/CN=classified.apps.example.com The application is reachable only at the following address:

Solution:

```
oc new-project area51
```

```
oc new-app --name=oxcart --image=quay.io/redhattraining/hello-world-nginx
```

```
openssl req -x509 -sha256 -nodes -days 365 -newkey rsa:4096 -keyout private.key -out  
certificate.crt
```

```
oc get svc
```

```
oc create route edge oxcart --service=oxcart --cert=<.cert> --key=<.key> --  
hostname=oxcart.apps.ocp4.example.com
```

```
oc get route; curl -vk https://oxcart.apps.ocp4.example.com
```

Note: A script named newcert is already available to create the certificates and keys

```
[student@workstation ~]$ newcert
```

[illegible]

```
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
```

What you are about to enter is what is called a Distinguished Name or a DN.
 There are quite a few fields but you can leave some blank
 For some fields there will be a default value,
 If you enter '.', the field will be left blank.

 Country Name (2 letter code) [XX]:IN
 State or Province Name (full name) []:TN
 Locality Name (eg, city) [Default City]:CH
 Organization Name (eg, company) [Default Company Ltd]:OSE
 Organizational Unit Name (eg, section) []:LAB
 Common Name (eg, your name or your server's hostname) []:classified.apps.example.com
 Email Address []:
 [student@workstation ~]\$ ls -ltr
 total 32
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Videos
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Templates
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Public
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Pictures
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Music
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Downloads
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Documents
 drwxr-xr-x. 2 student student 6 Jan 23 2024 Desktop
 drwxr-xr-x. 4 student student 35 Mar 13 22:27 D0280
 -rw-r--r--. 1 student student 1859 Mar 13 22:48 exam.sh
 -rw-r--r--. 1 student student 150 Mar 13 22:49 values.yml
 -rw-r--r--. 1 student student 101 Mar 13 22:50 deny-all.yml
 -rw-r--r--. 1 student student 251 Mar 14 02:06 ex280-htpasswd
 -rw-r--r--. 1 student student 858 Mar 14 02:09 oauth.yml
 -rw-r--r--. 1 student student 401 Mar 14 03:01 limitrange.yml
 -rw-----. 1 student student 3272 Mar 14 03:10 private.key
 -rw-r--r--. 1 student student 2037 Mar 14 03:11 certificate.crt
 [student@workstation ~]\$

```
[student@workstation ~]$ oc get route
NAME      HOST/PORT          PATH      SERVICES  PORT      TERMINATION  WILDCARD
oxcart    oxcart.apps.ocp4.example.com  oxcart    8080-tcp   edge        None
[student@workstation ~]$ oc delete route oxcart
route.route.openshift.io "oxcart" deleted
[student@workstation ~]$ oc create route edge oxcart --service=oxcart --cert=certificate.crt
--key=private.key --hostname=oxcart.apps.ocp4.example.com
route.route.openshift.io/oxcart created
[student@workstation ~]$ oc get route
NAME      HOST/PORT          PATH      SERVICES  PORT      TERMINATION  WILDCARD
oxcart    oxcart.apps.ocp4.example.com  oxcart    8080-tcp   edge        None
[student@workstation ~]$ curl -k https://oxcart.apps.ocp4.example.com
<html>
  <body>
    <h1>Hello, world from nginx!</h1>
  </body>
</html>
[student@workstation ~]$
```

<https://oxcart.apps.example.com>

Q.7: Configure application data

Deploy an application using the openshift/hello-openshift or image that meets the following requirements;

The application is part of a project named: acid The application is named: phosphoric The application uses a key named RESPONSE in a configuration map named sedicen The application is running and available at <http://phosphoricacid.apps.example.com> and display

“Soda pop won't stop can't stop”

Solution:

```
oc new-project acid
```

```
oc new-app --name=phosphoric --image=quay.io/redhattraining/hello-openshift
```

```
oc whoami; oc expose svc phosphoric --hostname=phosphoricacid.apps.ocp4.example.com
```

```
oc create configmap sedicen --from-literal=RESPONSE="Soda pop won't stop can't stop"
```

```
oc set env deployment/phosphoric --from configmap/sedicen
```

```
oc get route; curl -k http://phosphoricacid.apps.ocp4.example.com
```

Q. 8: Deploy an application

Deploy the chart named redhat-movie in the project ascii-movie from the repository <http://helm.domain20.example.com/chart/> or <http://helm.ocp4.example.com/charts> You may use the telnet commands to validate the deployment.

Solution:

```
oc new-project ascii-movie
```

```
helm repo add ex280 http://helm.ocp4.example.com/charts
```

```
helm search repo ex280
```

```
helm install redhat-movie ex280/etherpad
```

In a LAB: (For Reference)

```
helm repo add ex280 http://helm.ocp4.example.com/charts
```

```
helm search repo ex280
```

```
helm install example-app do280-repo/etherpad -f values.yaml --version 0.0.6
```

create a values.yaml as below;

image:

repository: registry.ocp4.example.com:8443/etherpad

name: etherpad

tag: 1.8.18

route:

host: development-etherpad.apps.ocp4.example.com

Q:9: Configure a secret

Configure a secret in the math project with the following requirements: The name of the secret is: magic.

The secret defines a key with name: decoder-ring The secret defines the key with value: XpWy9KdcP3T

Configure an application to use a secret Configure the application called qed in the math project with the following requirements: The application uses the secret previously created called: magic The secret defines an environment variable with name: decoder_ring The application output no longer displays: Sorry, application is not configured correctly.

Solution:

```
oc new-project math
```

```
oc create secret generic magic --from-literal=decoder-ring=XpWy9KdcP3T
```

```
oc get secret magic
```

```
oc new-app --name=qed --image=quay.io/redhattraining/hello-world-nginx
```

```
oc expose svc qed
```

```
oc get deploy
```

```
oc get secret magic
```

```
oc set env deployment/qed --from secret/magic
```

```
oc get route; curl -k http://route
```

Q: 10: Configure a service account

Configure a service account in the apples project to meet the following requirements: **ex280sa** The service account allows pods to be run as any available user.

Solution:

```
oc new-project apples
```

```
oc create serviceaccount ex280sa
```

```
oc adm policy add-scc-to-user anyuid -z ex280sa
```

```
oc get sa
```

Q: 11: Deploy an application

Deploy the application called oranges in the apples project so that the following condition No configuration components have been added or removed The application produces output The application uses the **ex280sa** service account

```
oc project apples
```

```
oc new-app --name=oranges --image=quay.io/redhattraining/gitlab-ce:8.4.3-ce.0
```

```
oc expose svc oranges
```

```
oc get sa
```

```
oc set sa deployment/oranges ex280sa
```

```
oc get route; curl -k http://route ; here output is not working since deployment name is wrong in service selector.
```

```
oc edit svc oranges; updated the deployment name properly in selector
```

```
oc get pod; curl -k http://route and now it should work .
```

Q:12: Deploy an application

Deploy the application called atlas in the mercury project so that the following condition should met. No configuration components have been added or removed The application produces output.

Solution:

```
oc new-project mercury
```

```
oc new-app --name=atlas --image=quay.io/redhattraining/hello-world-nginx
```

```
oc expose svc atlas
```

```
oc set resources deployment/atlas --requests=cpu=25m --limits=cpu=180m
```

```
oc edit deployment atlas
```

go to the spec:

Containers:

here make sure to remove existing memory and update as resources: {}

---save---

```
oc get deploy,pod
```

```
oc get route
```

curl -k http://route; it should work now

Q:13: Configure a network policy. (More Weightage)

Configure a network policy using the database and checker projects with the following requirements:

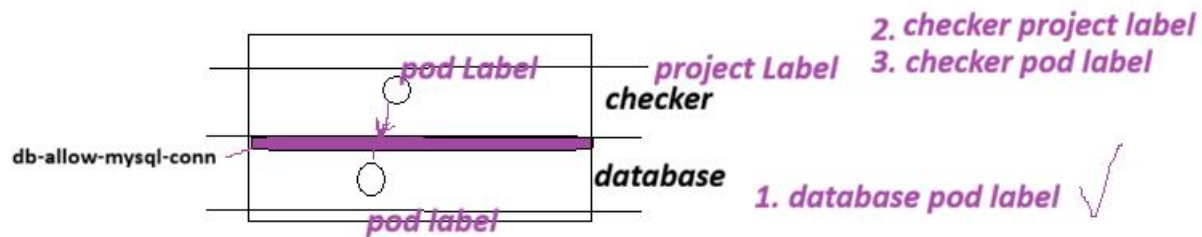
The database project has network policy with the name **db-allow-mysql-conn** based on pod selector label `network.openshift.io/policy-group` Connections to the database project are restricted to deployments from the checker project The network policy is filtered by project selector using the

`team:devsecops` label and pod selector using the `deploymentdb-mysql` label

The application can establish a connection to port 3306/TCP

You can check your work by examining the logs in the checker project.

we can create the resources for a lab with help of given files



```
[student@workstation ~]$ oc get deploy --show-labels -n database
NAME          READY  UP-TO-DATE  AVAILABLE  AGE  LABELS
mydatabase    1/1    1            1           11m  app.kubernetes.io/component=mydatabase,app.kubernetes.io/instance=mydatabase,app=mydatabase,network.openshift.io/policy-group=mysql-db
[student@workstation ~]$ oc get project checker --show-labels
NAME      DISPLAY NAME  STATUS  LABELS
checker   Active        kubernet
es.io/metadata.name=checker,pod-security.kubernetes.io/audit-version=latest,pod-security.kubernetes.io/audit=restricted,pod-security.kubernetes.io/warn-version=latest,pod-security.kubernetes.io/warn=restricted,team=devsecops
[student@workstation ~]$ oc get deploy -n checker --show-labels
NAME          READY  UP-TO-DATE  AVAILABLE  AGE  LABELS
checker-client 1/1    1            1           3m46s  app.kubernetes.io/component=checker-client,app.kubernetes.io/instance=checker-client,app=checker-client,deployment-db=mysql
[student@workstation ~]$
```

Solution:

we can create the resources for a lab with help of given files

steps:

oc project database; oc get pod --show-labels

oc project checker; oc get project checker --show-labels; oc get pod --show-labels

goto database project ==> choose network resource ==> create a networkpolicy ==> add the policy name ==> get the database pod label=selector and update here ==> add Ingress ==> Ingress rule to choose as Allow pods from inside the cluster ==> get the checker project label=selector and get the checker pod label=selector ==> Ports where update the 3306 with TCP and create

os project checker; oc rsh pod/podname; telnet databaseip:3306

Q-14: Configure a project template . (More Weightage)

Configure your OpenShift cluster so that new projects are created with limits using the following requirements

The name of the limit range is: PROJECT_NAME-Limits where PROJECT_NAME is the name of the project created using `oc new-project`

The amount of memory consumed by a single container is between 256Mi and 512Mi with a default of 512M and a default request of 256Mi

Solution:

```
#oc adm create-bootstrap-project-template -o yaml > template.yaml
```

```
#vim template.yaml
```

append the below limitrange resource

```
apiVersion: v1
```

```
kind: LimitRange
```

```
metadata:
```

```
  name: ${PROJECT_NAME}-limits
```

```
  namespace: ${PROJECT_NAME}
```

```
spec:
```

```
  limits:
```

```
    - type: Container
```

```
      max:
```

```
        memory: "512Mi"
```

```
      min:
```

```
        memory: "256Mi"
```

```
      default:
```

```
        memory: "512Mi"
```

```
      defaultRequest:
```

```
        memory: "256Mi"
```

```
:wq
```

```
oc create -f template.yaml -n openshift-config
```

oc edit projects.config.openshift.io cluster

```
spec:
```

```
  projectRequestTemplate:
```

```
    name: project-request
```

```
:wq
```

```
oc get pod -n openshift-apiserver
```

as part of validation Once the new pod gets created properly then create a new project and validate the limit range

```
oc new-project; oc get limitranges; oc describe limitranges project-requestor
```

Q: 15: Install the file-integrity operator with the following The operator is installed in the openshift-file-integrity project The approval strategy is Automatic Cluster monitoring is enabled for the openshift file-integrity project

Solution:

Goto operator and choose the requested operator and install

make sure to tick the check box and make sure to expected namespace and install it and then verify

Q:16: Create a cron job using the image at registry.domain20.example.com/library/job-runner:latest with the following requirements: The cron job name is job-runner The cron job runs at 04:05 on the 2nd day of every month The successful job history limit is 14 The service account and service account name is magna The cron job runs in the project called elementum

Solution:

oc new-project elementum

```
oc create cronjob job-runner --image=quay.io/redhattraining/hello-world-nginx --schedule='05 04 02 * *'
```

```
oc describe cronjobs.batch
```

```
oc create serviceaccount magna
```

```
oc get cronjobs.batch job-runner
```

```
oc edit cronjobs.batch job-runner
```

spec:

Container:

--

here we need to add the serviceAccountName:magna and successful job history limit is 14

```
:wq
```

```
oc get cronjobs.batch job-runner -o yaml |grep -i serviceAccount -A10
```